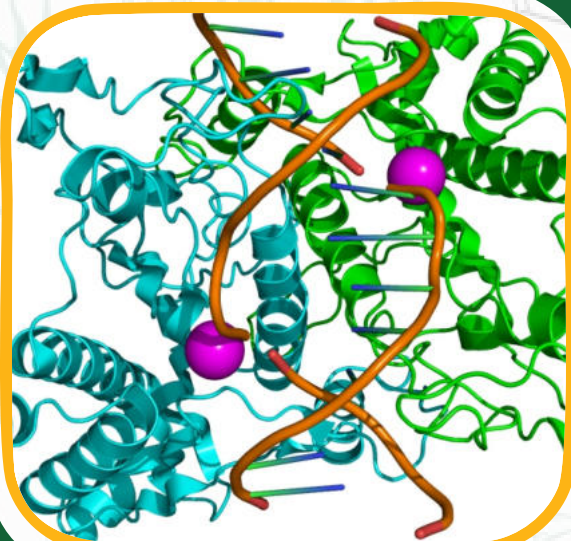
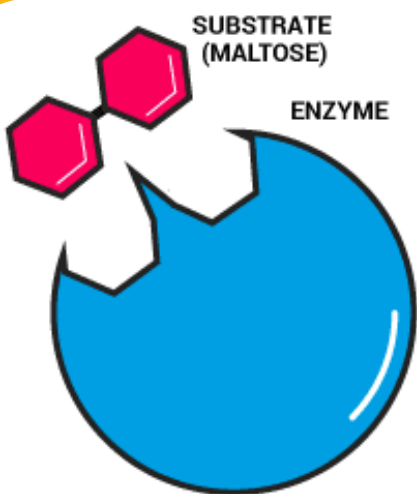




ENZYMES



DEFINITIONS:**ENZYMES:**

Enzymes are the biocatalyst which facilitate chemical reaction by lowering activation energy.

METABOLIC REACTION:

The reactions of an organisms are collectively called metabolic reactions.

METABOLISM:

The phenomena of chemical activity called metabolism.

ACTIVE SITES:

A small portion of enzymes where substrate attaches with enzyme is called active site.

SHORT/DETAILED ANSWER & QUESTIONS**Q. Define Metabolism.**

Ans. **METABOLISM**

DEFINITION:

"Metabolism is a set of biochemical reactions that occur in living organisms in order to maintain life."

These processes allow organisms to grow and reproduce, maintain their structure and respond to their environments.

Q. What is the difference between catabolism and anabolism?

Ans. **DIFFERENCE BETWEEN CATABOLISM AND ANABOLISM**

CATABOLISM	ANABOLISM
Catabolism includes the biochemical reactions in which large molecules are broken down.	Anabolism includes the biochemical reactions in which larger molecules are synthesized.
Energy is released.	Energy is utilized.
Example, Cellular respiration.	Example, Photosynthesis.

Q. What are Enzymes? Describe characteristics of enzymes.

Ans. **ENZYMES:**

DEFINITION:

"Enzymes are proteins that catalyze (i.e., speed up) biochemical reactions and are not changed during the reaction."

During metabolism, chemical is transformed from one form to other by enzymes. They act as biocatalyst and speed and regulate metabolic activity.

CHARACTERISTICS OF ENZYMES:

- i. When an enzyme starts a chemical reaction, catalyzes the reaction hence does not utilized itself which means even a single or little number of enzymes can start a reaction and catalyze fastly.
- ii. Their presence does not affect the nature or properties of end products.
- iii. Reactants of enzymes are called substrate.
- iv. They are very specific in their action; a single enzyme catalyzes only a single chemical reaction or a group of related reactions.

- v.** A small portion of enzymes where substrate attaches with enzymes is called Active site. The shape of active site is complementary to shape of substrate.
- vi.** They are sensitive to even a minor change in pH, temperature and substrate concentration.
- vii.** Some enzymes require cofactor for their functioning; a cofactor is a non-protein substance which may be organic or inorganic. Z^{+2} , Mg^{+2} , Mn^{+2} , Fe^{+2} , Cu^{+2} , K^{+1} and Na^{+1} the organic cofactors are NADD, NAD and FAD are used in enzymes as cofactors
- viii.** Many enzymes work in a sequential manner to produce a specific product. This pathway is called metabolic pathway.
- ix.** Activity of enzymes can be enhanced by activator and can decreased by inhibitors.
- x.** An enzyme inhibitor is a molecule that binds to an enzyme and decreases its activity. Since blocking an enzymes activity can kill a pathogen.

Q. Write the uses of enzymes.

Ans. **USES OF ENZYMES:**

Many enzymes are used commercially in industries. The most common industries are.

1. PAPER INDUSTRY:

To get cellulose for paper making.

2. FOOD INDUSTRY:

For making Bakery products and pizza.

3. BREWING INDUSTRY:

For conversion of sugar into alcohol.

4. BIO DETERGENTS:

Used to remove different type of stains.

Q. Describe factors affecting enzyme activity.

Ans. **FACTORS AFFECTING THE ACTIVITY OF AN ENZYME:**

Enzymes are very sensitive to the environment in which they work. The activity of an enzyme is affected by any change in condition that alters its chemistry and its shape. Some of the factors that can affect the rate of enzyme action are being discussed below.

1. TEMPERATURE:

Increase in temperature speed up the rate of enzyme catalyzed reactions, but only to a point.

OPTIMUM TEMPERATURE:

Every enzyme works at its maximum rate at a specific temperature called as the optimum temperature of that enzyme. When temperature rises to a certain limit, the heat adds in the activation energy and also provides kinetic energy and so reaction are accelerated. When temperature is raised well above the optimum temperature, the heat energy increases the vibrations of atom of enzyme and the globular structure of enzyme is lost. This is known as denaturation of enzyme.

It results in a rapid decrease in the rate of enzyme action and it may be blocked completely.

2. SUBSTRATE CONCENTRATION:

If there are enzymes molecules with vacant sites, an increase in substrate concentration will increase the rate of reaction.

SATURATION OF ACTIVE SITE:

If the enzymes concentration is kept constant and the amount of substrate is increased, a point is reached where any further increase in substrate does not increase the rate of reaction any more. When all the active sites of the enzymes are occupied at (high substrate concentration),

any more substrate molecules do not find free active sites. This state is called saturation of active sites and does not increase the rate of reaction.

3. pH (OPTIMUM pH):

All enzymes work at their maximum rate at a narrow range of pH, called as the optimum pH. A slightly changes (increase or decrease) in this pH causes retardation in enzyme activity or blocks its completely.

Every enzyme has its specific optimum pH values for example,

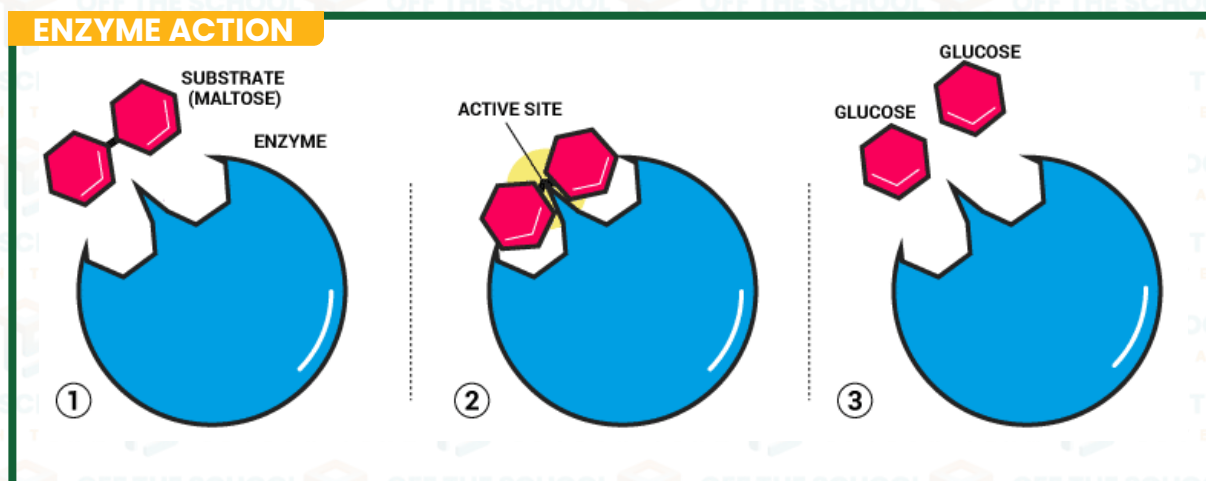
- Pepsin (working in stomach) is active in acidic medium (low pH).
- trypsin (working in small intestine) shows its activity in alkaline/basic medium (high pH).

Change in pH can affect the ionization of the amino acids at the active sites.

Q. What is the difference between catabolism and anabolism?

Ans. **MECHANISMS OF ENZYME ACTION:**

Enzyme catalysis the reaction by attaching to substrate which ends to the product formation. Enzymes exposes its active site to attract specific substrate, make enzyme substrate complex (ESC) after which the product is formed and enzyme is detached from it and used again for the same reaction.

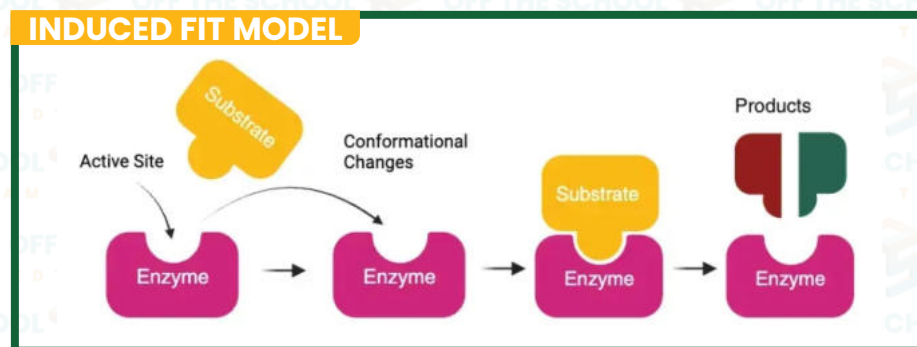
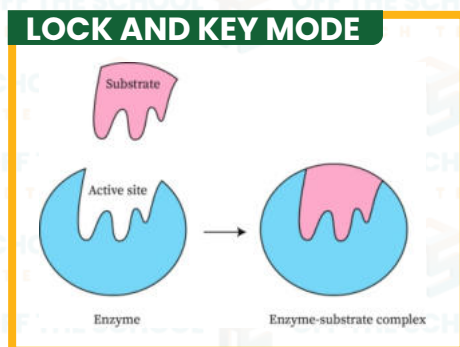


LOCK AND KEY MODEL:

In order to explain the mechanism of enzyme action, a German chemist Emil Fischer, in 1894 proposed the lock and key model. According to this model, both enzyme and the substrate possess specific shapes that fit exactly into one another. This model explains enzyme specificity.

INDUCED FIT MODEL:

The induced fit model suggested by Daniel Kohled in 1958, it explains that active site continuously changes its shape until the substrate bind to it. It also said that active site of enzyme is flexible (lock and key theory does not explain it.)



Q. Describe the specificity of enzymes?

Ans. **ACTIVATION ENERGY:**

The chemical reaction requires particular conditions to carry down at proper rate, temperature and pressure. The condition of temperature and pressure inside cell or organism are generally found not suitable for chemical reaction e.g., inside human body normal temperature remain 37°C and pressure is 120/80m.m of Hg. These conditions of temperature and pressure are not enough to perform any chemical reactions. Now body requires some facilitators. These facilitators help to perform biochemical reactions at low energy. it is clear now that each reaction requires some amount of minimum energy to initiate a reaction. This minimum amount of energy is called activation energy. If this amount is high, the difficult will be the reaction of vice versa e.g., the activation energy needed to break a glucose molecule initially requires energy of 2 ATP molecules

ENZYMES

